

Experimental Design Practical

Contents

1. Human practical: Identification of prognostic biomarkers in human prostate cancer patients 2
2. Mouse practical (Hypothetical example): Elucidating the Transcriptomic Landscape and Signaling Pathways Regulated by Hyperactive HHEX During Murine Organogenesis 3

1. Human practical: Identification of prognostic biomarkers in human prostate cancer patients

A biomarker is any measurable biological characteristic that serves as an indicator of a specific disease state, reflecting its presence, progression, or severity. In oncology, biomarkers are invaluable clinical tools that can be non-invasively or minimally invasively quantified from a variety of biological matrices, including blood, urine, saliva, or solid tissue biopsies.

Specifically, prognostic biomarkers evaluate a patient's baseline risk, providing a statistical forecast of the likely course of the disease (such as overall or recurrence-free survival) independent of any specific therapeutic intervention.

Study aim:

The aim of this study is to identify a panel of prognostic biomarkers that can predict recurrence-free survival in patients with prostate cancer. Such biomarkers could help identify patients at higher risk of disease recurrence who may benefit from more aggressive treatment, while sparing patients with a more favourable prognosis from unnecessary treatment-related side effects and overtreatment.

1. What would the ideal study cohort be?
2. What factors make above ideal cohort difficult to achieve?
3. In reality, what cohort would be chosen? What issues might this cause?
4. What are the pros and cons for using a retrospective or prospective cohort for this study? [A retrospective cohort study looks back over time and takes data from patients notes and/or interviews. A prospective cohort study follows the cohort through time collecting data as it goes].
5. What would the outcome measure(s) be for this study? What issues can arise in the definition of the outcome measures for this study?
6. What factors might affect the outcome? What information may you need to collect for use in the study analysis?
7. What are your groups? Do you need a control group in addition? If so what will your control group be?
8. What effect would a short follow-up time have? Which group will this affect more?
9. Which technologies could you use to screen for your panel of biomarkers?
10. What factors might influence your choice of technology for this study?
11. Once your study has a panel of biomarkers from which a test for recurrence has been formulated, what is the next step?
12. What characteristics would a clinical test formed from your biomarkers need?

2. Mouse practical (Hypothetical example): Elucidating the Transcriptomic Landscape and Signaling Pathways Regulated by Hyperactive HHEX During Murine Organogenesis

The homeobox transcription factor HHEX is a critical orchestrator of murine brain and liver development. While wild-type HHEX undergoes precise regulatory control, a mutant variant—characterized by the substitution of all Serine and Tyrosine residues with Alanine—exhibits constitutive hyperactivity, resulting in embryonic lethality in homozygous mice. This study aims to delineate the downstream molecular mechanisms and aberrant gene regulatory networks driven by HHEX hyperactivity. Using embryonic day 15 (E15) fetuses, we will isolate brain and liver tissues across three distinct morphological genotypes: homozygous wild-type ($HHEX^{Wt/Wt}$), heterozygous mutant ($HHEX^{Wt/Mt}$), and homozygous mutant ($HHEX^{Mt/Mt}$). Through comparative transcriptomic profiling, we seek to identify global gene expression alterations and map the key signaling pathways disrupted by hyperactive HHEX, providing deeper insights into its role as a master developmental regulator.

Experimental design related question and answers:

- 1) What are the scientific questions of interest in this experiment?
- 2) What are the independent and dependent variables in this experiment?
- 3) What are the null and alternative hypotheses in this case?
- 4) What control samples should be included in this experiment, and why are controls necessary?
- 5) How many replicates you need to include for each group? Discuss what factors might have influence in selecting the number of replicates?
- 6) Which experimental groups will be included in the study?
- 7) Assuming four replicates per group, calculate the total number of mice needed for this experiment. Consider two possible approaches: collecting both brain and liver tissue from the same animal, or using dedicated mice for each tissue type. Which approach would you recommend, and what is your reasoning?
- 8) How will any findings be validated?
- 9) What are possible sources of bias and confounding variables in the experiment?
- 10) How can these sources of bias and confounding be controlled?